

A
Major Project
on

ENHANCING PREDICTIVE ACCURACY FOR AGRICULTURAL CROP YIELDS IN INDIAN STATES

Submitted in partial fulfillment of the requirements for the award of degree

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted By

S.PRIYANKA	217Z1A05G7
MD ALTHAF PASHA	227Z5A0514
V.VIJAY	227Z5A0520

Under the Guidance of
Dr . K. SRINIVAS BABU

Professor



SCHOOL OF ENGINEERING
Department of Computer Science and Engineering

**NALLA NARASIMHA REDDY
EDUCATION SOCIETY'S GROUP OF INSTITUTION
(AN AUTONOMOUS INSTITUTION)**

**Approved by AICTE, New Delhi, Chowdariguda (V) Korremula 'x' Roads,
Via Narapally, Ghatkesar (Mandal) Medchal (Dist), Telangana-500088
2024-2025**



NALLA NARASIMHA REDDY
Education Society's Group of Institutions - Integrated Campus
(UGC AUTONOMOUS INSTITUTION)



SCHOOL OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the project report titled “**ENHANCING PREDICTIVE ACCURACY FOR AGRICULTURAL CROP YIELDS IN INDIAN STATES**” is being submitted by **S. PRIYANKA (217Z1A05G7) and MD. ALTHAF PASHA (227Z5A0514), V. VIJAY (227Z5A0520)** in Partial fulfilment for the award of **Bachelor of technology in Computer Science & Engineering** is a record bonfire work carried out by them. The results embodied in this report have not been submitted to any other University for the award of any degree.

Internal Guide

(Dr. K. Srinivas Babu)

Head of the Department

(Dr. K. Rameshwaraiah)

Submitted for Viva voce Examination held on

External Examiner

DECLARATION

We S.Priyanka, MD.Althaf Pasha and V.Vijay, the students of **Bachelor of Technology in Computer Science and Engineering, Nalla Narasimha Reddy Education Society's Group Of Institutions**, Hyderabad, Telangana, hereby declare that the work presented in this project work entitled **“ENHANCING PREDICTIVE ACCURACY FOR AGRICULTURAL CROP YIELDS IN INDIAN STATES”** is the outcome of our own bonafide work and is correct to the best of our knowledge and this work has been undertakentaking care of engineering ethics. It contains no material previously published or written byanother person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning.

By:

S. PRIYANKA	217Z1A05G7
MD ALTHAF PASHA	227Z5A0514
V. VIJAY	227Z5A0520

Date:

Signature:

ACKNOWLEDGEMENT

We express our sincere gratitude to our guide **Dr. K. Srinivas Babu**, Professor, in Computer Science and Engineering Department, NNRESGI, who motivated throughout the period of the project and also for his valuable and intellectual suggestions apart from hisadequate guidance, constant encouragement right throughout our work.

We would like to express our profound gratitude to our major project in-charge **Mrs.Ch. Ramya**, Assistant Professor, NNRESGI, for her support and guidance in completing our project and for giving us this opportunity to present the project work.

We profoundly express our sincere thanks to **Dr. K. Rameshwaraiah**, Professor & Head, Department of Computer Science and Engineering, NNRESGI, for his cooperation and encouragement in completing the project successfully.

We wish to express our sincere thanks to **Dr. G. Janardhana Raju**, Dean School ofEngineering, NNRESGI, for providing the facilities for completion of the project.

We wish to express our sincere thanks to **Dr. C. V. Krishna Reddy**, Director NNRESGI for providing the facilities for completion of the project.

Finally, we would like to thank Project Co-Ordinator, Project Review Committee (PRC) members, all the faculty members and supporting staff, Department of Computer Science and Engineering, NNRESGI for extending their help in all circumstances.

By:

S. PRIYANKA	217Z1A05G7
MD ALTHAF PASHA	227Z5A0514
V. VIJAY	227Z5A0520

ABSTRACT

Abstract—Agricultural productivity is critical for the sustenance and economic growth of India, a country with diverse climatic regions and crop practices. Accurate prediction of crop yields is essential for optimizing agricultural outputs, ensuring food security, and planning resource allocation. Traditional predictive models often struggle with the non-linear and complex nature of factors influencing crop yields. This study introduces an innovative approach by integrating power transformations, specifically the Yeo-Johnson transformation, within machine learning models to enhance the predictive accuracy of agricultural crop yields in various Indian states. We collected a comprehensive dataset comprising climatic variables, soil properties, crop types, and historical yield data across different Indian states. The Yeo-Johnson power transformation was applied to normalize the distribution of these features, addressing issues of skewness and heteroscedasticity, thereby making the data more suitable for machine learning algorithms. We evaluated several machine learning models, including Linear Regression, Random Forest, and Gradient Boosting, to establish a benchmark for comparison. Our findings demonstrate a significant improvement in predictive accuracy with the application of the Yeo-Johnson transformation across all models tested are shown in table 1. The Gradient Boosting model, post-transformation, exhibited the highest accuracy, underscoring the potential of combining power transformations with ensemble learning techniques for crop yield prediction. The study emphasizes the role of advanced data preprocessing techniques, such as power transformations, in unlocking the full potential of machine learning for agricultural applications. By enhancing the predictive accuracy of crop yield models, stakeholders can make more informed decisions, leading to improved agricultural productivity and sustainability in the diverse and challenging landscape of Indian agriculture.

TABLE OF CONTENTS

	Page No.
Abstract	
List Of Figures	i
List Of Tables	ii
1. INTRODUCTION	1
1.1 Motivation	1
1.2 Problem Statement	1
1.3 Purpose	1
1.4 Scope	1
1.5 Project Objective	2
1.6 Limitations	2
2. LITERATURE SURVEY	4
2.1 Introduction	4
2.2 Existing System	4
2.3 Proposed System	4
3. SYSTEM ANALYSIS	5
3.1 Functional Requirements	5
3.2 Non Functional Requirements	6
3.3 Interface Requirements	7

4. SYSTEM DESIGN	8
4.1 Uml Diagrams	8
4.2 Data Flow Diagram	11
4.3 Usecase Diagram	12
4.4 Class Diagram	13
4.5 Sequence Diagram	14
4.6 Activity Diagram	15
4.7 Modules	16
5. IMPLEMENTATION&RESULTS	17
5.1 Method of Implementation	17
5.2 Explanation of Key Functions	18
5.3 Output Screens	26
6. SYSTEM TESTING	33
6.1 Introduction	32
6.2 Various Test Cases Scenarios	33
7. CONCLUSIONANDFUTURE ENHANCEMENTS	36
7.1 Project Conclusion	36
7.2 Future Enhancements	36
8. REFERENCES	37
8.1 Journals	37
8.2 Web Links	38
8.3 Text Books	38

LIST OF FIGURES

Figure No.	Name Of The Figure	Page No.
4.1	DFD	11
4.2	Use Case Diagram	12
4.3	Class Diagram	13
4.4	Sequence diagram	14
4.5	Activity diagram	15
5.1	Home page	26
5.2	Admin Login	26
5.3	User Login	27
5.4	User Registration Form	27
5.5	Admin Home page	28
5.6	User List	28
5.7	Raw DataSet	29
5.8	Preprocesses Dataset	29
5.9	Reports	30
5.10	Results	30
5.11	Class Page	31
5.12	Class Details	31
5.13	Resource Optimazior	32

LIST OF TABLES

Table No.	Name Of The Table	Page No.
6.1	Test Cases	34-35

1. INTRODUCTION

1.1 MOTIVATION

By harnessing the power of machine learning, this project aims to revolutionize crop yield prediction in Indian agriculture. With accurate forecasts, farmers can make informed decisions, optimize resources, and improve productivity, ultimately contributing to a more sustainable and food-secure future for the nation.

1.2 Problem Statement

Accurate prediction of crop yields in Indian agriculture is crucial for ensuring food security, optimizing resource allocation, and promoting sustainability. However, the complex and non-linear nature of factors influencing crop yields poses significant challenges. This project aims to address the problem of inaccurate crop yield predictions by developing a machine learning model that integrates power transformations to enhance predictive accuracy.

1.3 PURPOSE

The purpose of this project is to develop and evaluate a machine learning model that accurately predicts crop yields in Indian agriculture, utilizing power transformations to enhance predictive accuracy, and providing insights for informed decision-making to improve agricultural productivity and sustainability.

1.4 SCOPE

This project is focused on developing and evaluating machine learning models to predict crop yields in Indian agriculture, with a specific emphasis on integrating power transformations to enhance predictive accuracy. The scope includes collecting and preprocessing relevant data, analyzing the impact of power transformation, and identifying key factors influencing crop yields, all within the context of Indian agriculture.

1.5 PROJECT OBJECTIVE

The primary objective of this project is to develop and evaluate a machine learning model that accurately predicts crop yields in Indian agriculture, leveraging the Yeo-Johnson power transformation to enhance predictive accuracy. Additionally, the project aims to identify key factors influencing crop yields and provide actionable insights for stakeholders to improve agricultural productivity and sustainability.

1.6 LIMITATIONS OF PROJECT

- 1. Data quality and availability:** Limited access to comprehensive and accurate data on crop yields, weather patterns, and soil conditions.
- 2. Geographical constraints:** Focus on Indian agriculture might limit applicability to other regions or countries.
- 3. Crop specificity:** Model might not perform equally well for all crop types or varieties.
- 4. Seasonal variability:** Model's performance might be affected by unusual weather patterns or seasonal fluctuations.
- 5. Model interpretability:** Complex machine learning models can be difficult to interpret, making it challenging to understand underlying factors driving predictions.
- 6. Overfitting or underfitting:** Model might not generalize well to new, unseen data due to overfitting or underfitting.
- 7. Lack of real-time data:** Model's performance might be impacted by delayed or incomplete data.
- 8. Model generalizability:** The model's performance may vary across different agricultural regions, crops, and climatic conditions.
- 9. Complexity of agricultural systems:** Crop yields are influenced by numerous factors, including weather, soil, and management practices, which can be challenging to capture accurately.
- 10. Power transformation limitations:** The Yeo-Johnson transformation may not always be effective in stabilizing variance or improving model performance.

11. Computational resources: Limited computational resources can impact model complexity and scalability.

12. Evaluation metrics: Choosing inappropriate evaluation metrics might not accurately reflect model performance.

2. LITERATURE SURVEY

2.1 INTRODUCTION

Thomas van Klompenburg ^a, Ayalew Kassahun ^a, Cagatay Catal, Machine learning is an important decision support tool for crop yield prediction, including supporting decisions on what crops to grow and what to do during the growing season of the crops. Several machine learning algorithms have been applied to support crop yield prediction research. In this study, we performed a Systematic Literature Review (SLR) to extract and synthesize the algorithms and features that have been used in crop yield prediction studies. Based on our search criteria, we retrieved 567 relevant studies from six electronic databases, of which we have selected 50 studies for further analysis using inclusion and exclusion criteria. We investigated these selected studies carefully, analyzed the methods and features used, and provided suggestions for further research. According to our analysis, the most used features are temperature, rainfall, and soil type, and the most applied algorithm is Artificial Neural Networks in these models.

2.2 EXISTING SYSTEM

The existing crop yield prediction systems in India primarily consist of statistical models, such as linear regression and time-series analysis, expert-based systems that leverage local agricultural knowledge, remote sensing and GIS technologies that utilize satellite imagery and spatial analysis, and crop simulation models like DSSAT and APSIM. These systems provide valuable insights but are limited by their predictive precision, scalability, and adaptability to complex agricultural systems and climate change.

2.3 PROPOSED SYSTEM

The proposed model enhances crop yield prediction by integrating Yeo-Johnson power transformation with machine learning techniques. It collects and preprocesses diverse datasets, applies feature selection, and trains various machine learning models. The model undergoes hyperparameter tuning and is deployed on a user-friendly platform with a feedback loop for continuous improvement, providing accurate and actionable insights for stakeholders to improve agricultural productivity and sustainability.

3. SYSTEM ANALYSIS

3.1 FUNCTIONAL REQUIREMENTS

- **Data Collection:** Collect diverse datasets (weather, soil, historical yield)
- **Data Preprocessing:** The system should be able to clean, encode, and normalize the collected data to ensure high data quality.
- **Yeo-Johnson Transformation:** The system should be able to apply the Yeo-Johnson transformation to skewed or non-normally distributed features to normalize the data and stabilize.
- **Feature Selection:** The system should be able to select the most predictive variables and retain only those that contribute significantly to the model's performance.
- **Model Training:** The system should be able to train various machine learning models, including linear regression, tree-based methods, and neural networks.
- **Hyperparameter Tuning:** The system should be able to optimize the performance of the machine learning models through hyperparameter tuning.
- **Prediction:** The system should be able to generate accurate crop yield predictions based on the trained models.
- **Graphical User Interface (GUI):** The system should provide a user-friendly GUI that allows users to interact with the system, input data, view predictions, and provide feedback.

3.2 NON-FUNCTIONAL REQUIREMENTS

- **Performance:** The system should be able to generate accurate predictions in a timely manner.
- **Scalability:** The system should be able to handle large datasets and user traffic without compromising performance.
- **Security:** The system should ensure the security and integrity of user data and prevent unauthorized access.
- **Usability:** The system should be easy to use and provide an intuitive interface for.
- **Maintainability:** The system should be easy to maintain and update, with minimal downtime
- **Reliability:** The system should provide consistent and reliable performance, with minimal errors.
- **Adaptability:** The system should be able to adapt to changing agricultural conditions and new data.

3.3 INTERFACE REQUIREMENTS

- **Form:** A form for users to input data, such as weather conditions, soil properties, and crop type.
- **Prediction Display:** A section to display the predicted crop yield and provide insights into the factors that influence the prediction.
- **Feedback Mechanism:** A mechanism for users to provide feedback on the accuracy of the predictions and suggest improvements.
- **Data Visualization:** Tools to visualize data, such as charts and graphs, to help users understand trends and patterns.

Hardware Requirements:

- System : Intel Core i7.
- Hard Disk : 1TB.
- Monitor : 15'' LED
- Input Devices : Keyboard, Mouse
- Ram : 16 GB.

Software Requirement:

- Operating system : Windows 11.
- Coding Language : Python.
- Front-End : Html. CSS
- Designing : Html ,css,javascript.
- Data Base : SQLite3

4.SYSTEM DESIGN

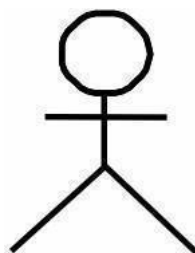
4.1 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

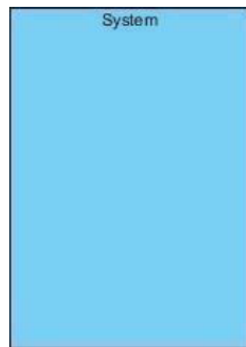
Actor: An actor is a model element that interacts with a system



Use Case: A use case describes a function that a system performs to achieve the user's goal



Use case system: A system of a use case defines and represents boundaries of a business, software system, physical system or device, subsystem,



Associations: A line between actors and use cases. It describes which actors are associated with which use cases



Class: A class represents an object or a set of objects that share a common structure and behavior.



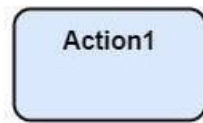
Generalization: A generalization is a relationship between a parent class and a child class. In this, the child class is inherited from the parent class.



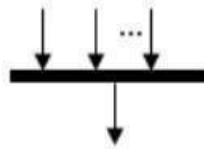
Composition: It portrays the dependency between the parent and its child



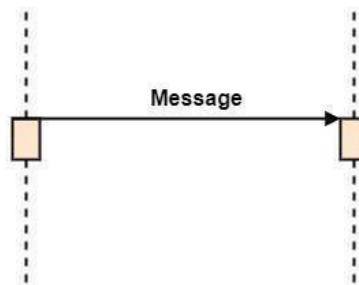
Action: An action is a named element that is the fundamental unit of an executable functionality.



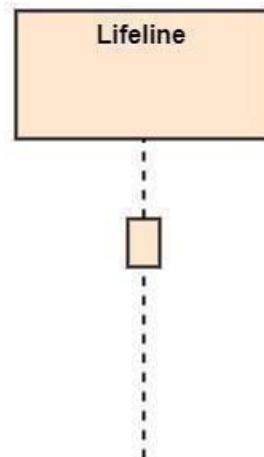
Join: Join nodes are used to support concurrent activities converging into one



Call Message: It depicts workflow or activity over time using messages passed from element to element



Activation: It is represented by a thin rectangle on the life line. It describes that time period in which an operation is performed by an element



Final State: It is the stage where all the control flows and object flow end.



4.2 Data flow Diagram

A data flow diagram (DFD) maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles and arrows, plus short text labels, to show data inputs, outputs, storage points and the routes between each destination. There are 2 levels in the below diagram.

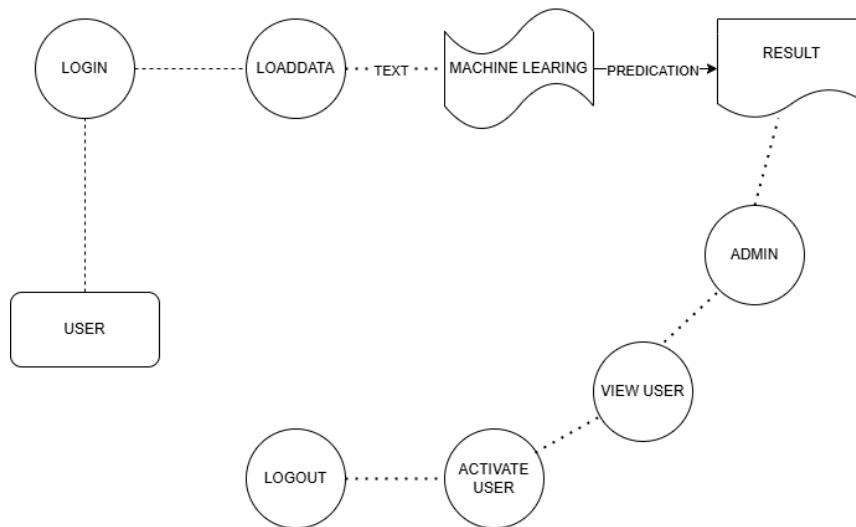


Fig :4.1 Data Flow Diagram

4.3. Use Case Diagram:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

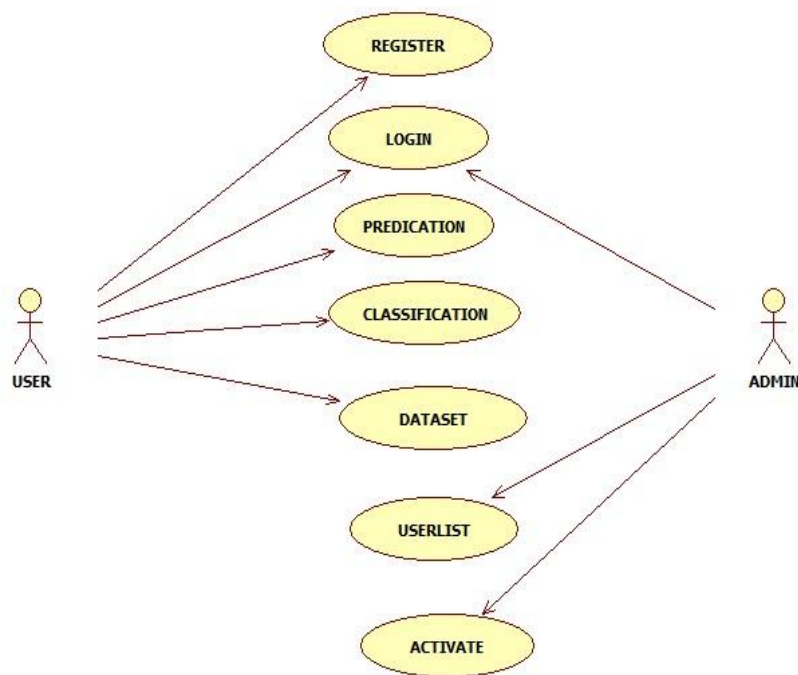


Fig :4.2 Use case Diagram

4.4. Class Diagram

software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

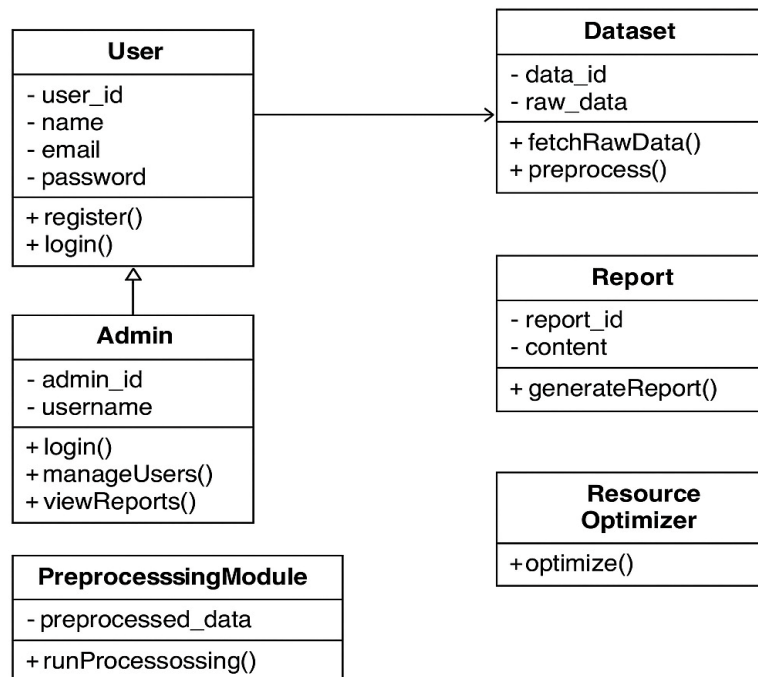


Fig :4.3 class diagram

4.5. Sequence Diagram:

A sequence diagram in Unified Modelling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

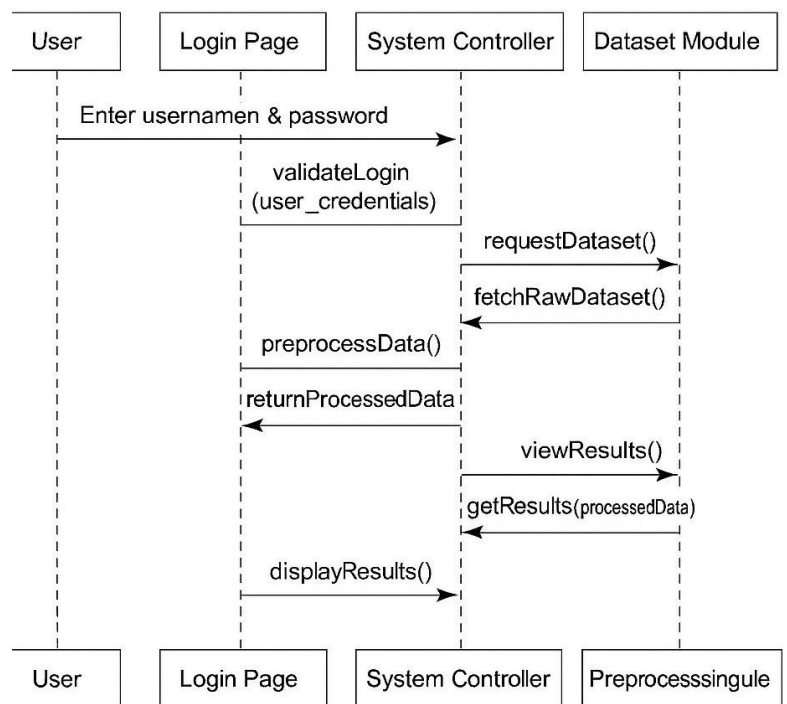


Fig :4.4 Sequence Diagram

4.6. Activity Diagram

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step- by-step workflows of components in a system. An activity diagram shows the overall flow of control.

Activity Diagram: Combined User/Admin Workflow

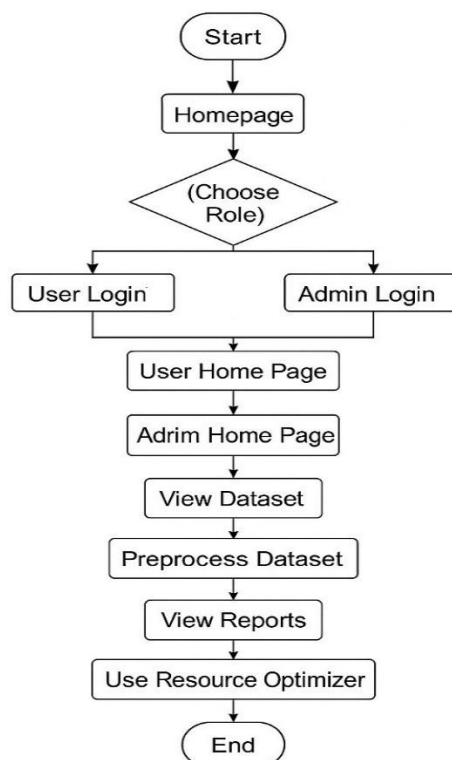


Fig :4.5 Activity Diagram

4.7. MODULES

USER:

The User can register the first. While registering he required a valid user email and mobile for further communications. Once the user register then admin can activate the user. Once admin activated the user then user can login into our system. User can upload the dataset based on our dataset column matched.

ADMIN:

Admin can login with his login details. Admin can activate the registered users. Once he activate then only the user can login into our system. Admin can view the overall data in the browser. Admin can click the Results in the web page so calculated Accuracy and macro avg and weighted avg based on the algorithms.

DATASET:

A **large dataset** in machine learning is one that contains a significant amount of data, often millions of data points. It plays a crucial role in improving model accuracy, generalization, and the ability to solve complex problems using advanced techniques like deep learning

REPORTS:

A confusion matrix is a table used to evaluate the performance of a classification model by comparing actual vs. predicted classifications. In binary classification, it consists of four key components: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

PREDICATIONS:

Predictions play a crucial role in academic projects, particularly in fields like data science, machine learning, and statistics, as they enable researchers to forecast future trends, outcomes, or behaviors based on existing data.

5. IMPLEMENTATION & RESULTS

5.1. MTHEOD OF IMPLIMENTATION

- **Step 1: Data Collection**
 - **1. Data Sources:** Identify and collect data from various sources, such as weather stations, soil sensors, and historical yield data.
 - **2. Data Format:** Ensure data is in a suitable format for analysis, such as CSV or JSON.
- **Step 2: Data Preprocessing**
 - **1. Data Cleaning:** Clean and preprocess data to handle missing values, outliers, and errors.
 - **2. Feature Engineering:** Extract relevant features from the data, such as weather patterns and soil properties.
- **Step 3: Model Development**
 - **1. Machine Learning Library:** Use a suitable machine learning library, such as scikit-learn or TensorFlow.
 - **2. Model Selection:** Select a suitable machine learning algorithm, such as linear regression or Random Forest.
- **Step 4: Model Deployment**
 - **Deployment Platform:** Deploy the model on a suitable platform, such as a web application or mobile app.
 - **API Development:** Develop an API to interact with the model and retrieve predictions.
- **Step 5: GUI Development**
 - **GUI Framework:** Use a suitable GUI framework, such as React or Angular.
 - **User Interface:** Design a user-friendly interface to input data, view predictions, and provide feedback.

5.2. EXPLANATION OF KEY FUNCTION

- **Function Purpose:** 1. **Data Collection:** Collecting diverse datasets, including weather conditions, soil properties, and historical yield data.
- **2. Yeo-Johnson Transformation:** Applying the Yeo-Johnson transformation to normalize skewed data and stabilize variance.
- **3. Feature Selection:** Selecting the most predictive variables to retain and enhance model interpretability.
- **4. Machine Learning Model Training:** Training various machine learning models, including linear regression, tree-based methods, and neural networks.
- **5. Hyperparameter Tuning:** Optimizing model performance through hyperparameter tuning.

Source Code

```
import React, from flask import Flask, render_template, request, redirect,
url_for,session
import sqlite3
import pandas as pd
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder

app = Flask(__name__)

#=====homepage modules start =====
@app.route('/')
def home():
    return render_template('index.html')

@app.route('/logout')
def logout():
    return render_template('index.html')

@app.route('/register', methods=['POST', 'GET'])
def register():
    msg = ""
    if request.method == 'POST':
        try:
            name = request.form['name']
            loginid = request.form['loginid']
            email = request.form['email']
            password = request.form['password']
```

```

branch = request.form['branch']
collagename = request.form['collagename']
phone = request.form['phone']
locality = request.form['locality']
state = request.form['state']

status = 'waiting'
with sqlite3.connect('database.db') as con:
    cur = con.cursor()
    cur.execute("INSERT INTO students (name, loginid, email, password, branch,
collagename, phone, locality, state, status) VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?)", (name,
loginid, email, password, branch, collagename, phone, locality, state, status))
    con.commit()
    msg = "Record successfully added to database"
except Exception as e:
    con.rollback()
    msg = f"Error in the INSERT: {e}"
finally:
    con.close()
    return render_template('register.html', msg=msg)
return render_template('register.html')

```

```

@app.route('/adminlogin',methods=['POST','GET'])
def adminlogin():
    error = None
    if request.method == 'POST':
        loginid = request.form['loginid']
        password = request.form['password']
        if loginid == 'admin' and password == 'admin':
            return render_template('admins/adminhome.html')
        else:
            error = 'Invalid Credentials. Please try again.'
            return render_template('adminlogin.html',error=error)

```

```

@app.route('/userlogin', methods=['POST', 'GET'])
def userlogin():
    error = None
    if request.method == 'POST':
        loginid = request.form['loginid']
        password = request.form['password']
        with sqlite3.connect('database.db') as con:
            con.row_factory = sqlite3.Row
            cur = con.cursor()
            cur.execute("SELECT * FROM students WHERE name = ? AND password =
?", (loginid, password))
            user = cur.fetchone()
            if user:
                if user['status'] == 'waiting':
                    error = "Your account is not activated yet. Please contact admin."
                else:
                    return render_template('users/userhome.html')
            else:
                error = "Invalid credentials. Please try again."
        return render_template('userlogin.html', error=error)

```

#=====homepage modules ending =====

#=====admin panel start here =====

```

@app.route('/adminhome')
def adminhome():
    return render_template('admins/adminhome.html')

```

```

@app.route('/list')
def list():
    con = sqlite3.connect("database.db")
    con.row_factory = sqlite3.Row
    cur = con.cursor()

```

```

        return redirect(url_for('list'))

    else:
        with sqlite3.connect('database.db') as con:
            con.row_factory = sqlite3.Row
            cur = con.cursor()
            cur.execute("SELECT * FROM students WHERE rowid = ?", (uid,))
            user = cur.fetchone()
            return render_template('admins/edit_user.html', user=user, uid=uid)

#=====admin panel end =====

#=====user panel start =====

@app.route('/userhome')
def userhome():
    return render_template('users/userhome.html')

#=====user panel start =====

@app.route('/dataset')
def dataset():
    dataset_path = os.path.join(app.root_path, 'media', 'crop_production.csv')
    data = pd.read_csv(dataset_path,nrows=100)
    return render_template('users/dataview.html',data=data.to_html())

@app.route('/dataset1')
def dataset1():
    dataset_path = os.path.join(app.root_path, 'media', 'dani.csv')
    data = pd.read_csv(dataset_path,nrows=100)
    return render_template('users/dataview1.html',data=data.to_html())

@app.route('/predication')
def predication():
    dataset_path = os.path.join(app.root_path, 'media', 'crop_production.csv')

```

```

df = pd.read_csv(dataset_path)
df = df.drop(columns=['District_Name'])
label_encoder = LabelEncoder()
# Step 3: Apply label encoding to categorical columns
df['State_Name'] = label_encoder.fit_transform(df['State_Name'])
df['Season'] = label_encoder.fit_transform(df['Season'])
df['Crop'] = label_encoder.fit_transform(df['Crop'])
# Adding a new column Yield which indicates Production per unit Area.

df['Yield'] = (df['Production'] / df['Area'])
# Dropping missing values
df = df.dropna()
df.corr()
sns.heatmap(df.corr(), annot=True)
plt.title('Correlation Matrix')
dummy = pd.get_dummies(df)
dummy
from sklearn.model_selection import train_test_split
x = dummy.drop(["Production", "Yield"], axis=1)
y = dummy["Production"]
# Splitting data set - 25% test dataset and 75%
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.25, random_state=5)
# Training the Simple Linear Regression model .
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(x_train, y_train)
# Predicting the test Results
lr_predict = model.predict(x_test)
return render_template('users/predication1.html', accuracy1 = acc1 , accuracy=acc)
from joblib import load # Import joblib's load function

# Load your model from the media folder
model_path = os.path.join('media', 'random_forest_model1.pkl') # Adjust the model
filename

```

```

def connect_db():
    conn = sqlite3.connect('database.db')
    print("Connected to database successfully")

    # Check if the students table exists
    cursor = conn.cursor()
    cursor.execute("SELECT name FROM sqlite_master WHERE type='table' AND
name='students'")
    table_exists = cursor.fetchone()
    if not table_exists:
        # Create the students table if it doesn't exist
        conn.execute('CREATE TABLE students (name TEXT, loginid TEXT, email
TEXT, password TEXT, branch TEXT, collagename TEXT, phone TEXT, locality
TEXT, state TEXT, status TEXT)')
        print("Created table successfully!")
    else:
        print("Table 'students' already exists.")
    # Close the connection
    conn.close()

# Ensure the table is created before running the app
connect_db()

base.html
<!DOCTYPE html>
<html>
<head>
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}">
</head>
<body>
<div class="navbar-container">
    <div class="topnav">
        <a href="/">HOME</a>
        <a href="{{ url_for('userlogin') }}">USERLOGIN</a>
        <a href="{{ url_for('adminlogin') }}">ADMINLOGIN</a>

```

```

        <a href="{{ url_for('register') }}">USERREGISTERFORM</a>
    </div>
</div>
<!--  --
>
<main>
    {% block content %}

    {% endblock %}

</main>
<hr>
<!-- footer start -->
    <footer>
        <p>&copy; PROJECT DEVELOPED BY ALTHAF TEAM& </p>
    </footer>
<!-- endfooter -->
</body>
</html>

```

5.3 Output Screens



Fig: 5.1 Home page

The screenshot displays the Admin Login page of the web application. The browser address bar shows the URL 127.0.0.1:7000/adminlogin. The navigation bar includes links for HOME, USERLOGIN, ADMINLOGIN, and USERREGISTERFORM. The main content area features the "ADMIN LOGINFORM" with the following fields and buttons:

- Login ID:** A text input field with the placeholder "Enter User LoginId".
- Password:** A text input field with the placeholder "Enter User Password".
- Login:** A button to submit the login information.

There is also a small box in the top right corner showing "3.4 kbps" and "4.6 kbps".

Fig: 5.2 Admin Login

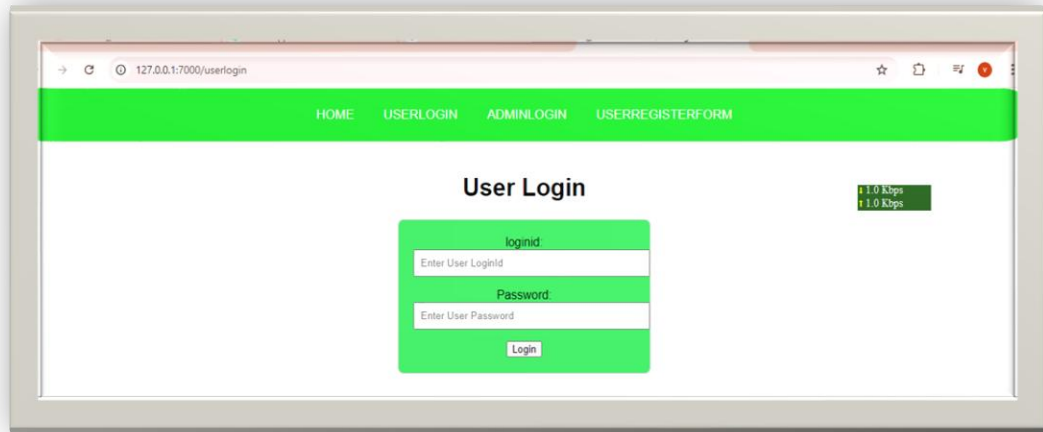


Fig:5.3 User login

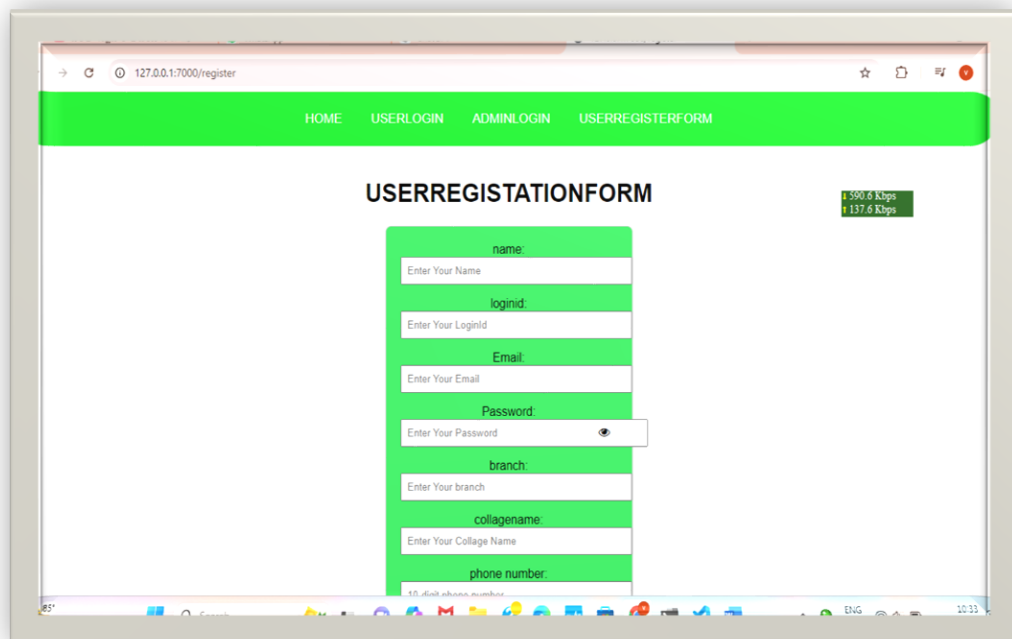


Fig:5.4 User Registration Form

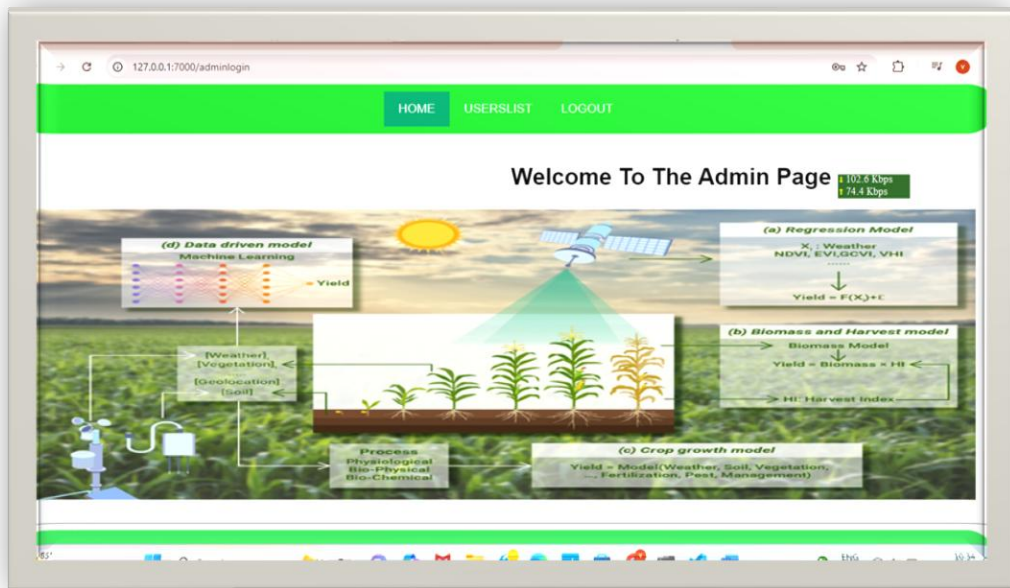


Fig 5.5 Admin Home Page

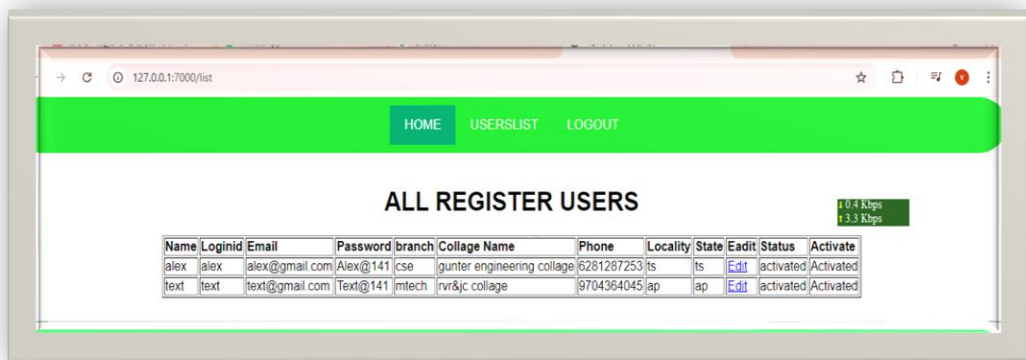


Fig 5.6 User List

DATASET

	State_Name	District_Name	Crop_Year	Season	Crop	Area	Production
0	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	Areca nut	1254.00	2000.00
1	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	Other Kharif pulses	2.00	1.00
2	Andaman and Nicobar Islands	NICOBARS	2000	Kharif	Rice	102.00	321.00
3	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	Banana	176.00	641.00
4	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	Cashewnut	720.00	165.00
5	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	Coconut	18168.00	65100000.00
6	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	Dry ginger	36.00	100.00
7	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	Sugarcane	1.00	2.00
8	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	Sweet potato	5.00	15.00
9	Andaman and Nicobar Islands	NICOBARS	2000	Whole Year	Tapioca	40.00	169.00
10	Andaman and Nicobar Islands	NICOBARS	2001	Kharif	Areca nut	1254.00	2061.00
11	Andaman and Nicobar Islands	NICOBARS	2001	Kharif	Other Kharif pulses	2.00	1.00
12	Andaman and Nicobar Islands	NICOBARS	2001	Kharif	Rice	83.00	300.00
13	Andaman and Nicobar Islands	NICOBARS	2001	Whole Year	Cashewnut	719.00	192.00
14	Andaman and Nicobar Islands	NICOBARS	2001	Whole Year	Coconut	18190.00	64430000.00
15	Andaman and Nicobar Islands	NICOBARS	2001	Whole Year	Dry ginger	46.00	100.00
16	Andaman and Nicobar Islands	NICOBARS	2001	Whole Year	Sugarcane	1.00	1.00
17	Andaman and Nicobar Islands	NICOBARS	2001	Whole Year	Sweet potato	11.00	33.00

Fig 5.7 Raw Data Set

PREPROCESSED DATASET

	State_Name	Crop_Year	Season	Crop	Area	Production	Yield
0	0	2000	1	2	1254.00	2000.00	1.594896
1	0	2000	1	74	2.00	1.00	0.500000
2	0	2000	1	95	102.00	321.00	3.147059
3	0	2000	4	7	176.00	641.00	3.642045
4	0	2000	4	22	720.00	165.00	0.229167
5	0	2000	4	28	18168.00	65100000.00	3583.223250
6	0	2000	4	38	36.00	100.00	2.777778
7	0	2000	4	106	1.00	2.00	2.000000
8	0	2000	4	108	5.00	15.00	3.000000
9	0	2000	4	109	40.00	169.00	4.225000
10	0	2001	1	2	1254.00	2061.00	1.643541
11	0	2001	1	74	2.00	1.00	0.500000
12	0	2001	1	95	83.00	300.00	3.614458
13	0	2001	4	22	719.00	192.00	0.267038
14	0	2001	4	28	18190.00	64430000.00	3542.056075
15	0	2001	4	38	46.00	100.00	2.173913
16	0	2001	4	106	1.00	1.00	1.000000
17	0	2001	4	108	11.00	33.00	3.000000

Fig 5.8 Preprocesses Dataset

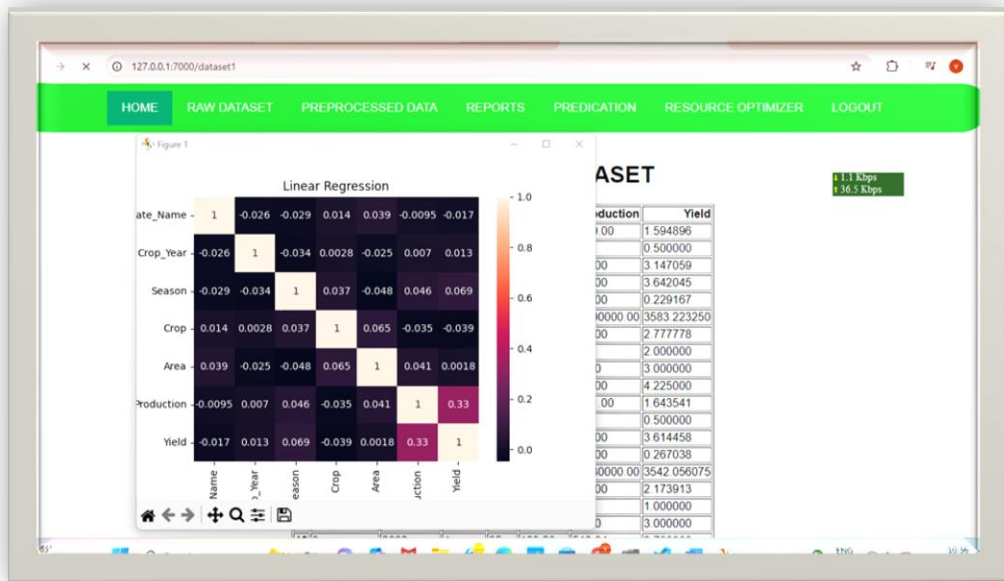


Fig 5.9 Reports

WELCOME TO YIELD PREDICATION:None per unit

State_Name: Select State_Name

Crop_Year: Enter Crop_Year

Season: Select Which Season

Crop: Select Which Crop

Area: Enter Area in units

submit

Fig 5.10 Result

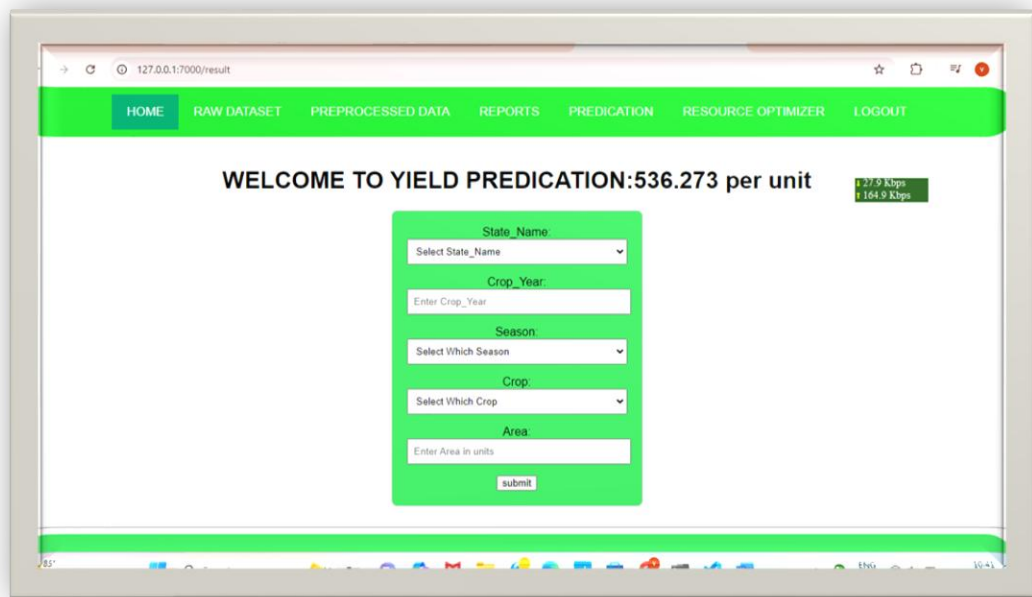


Fig 5.11 Class Page

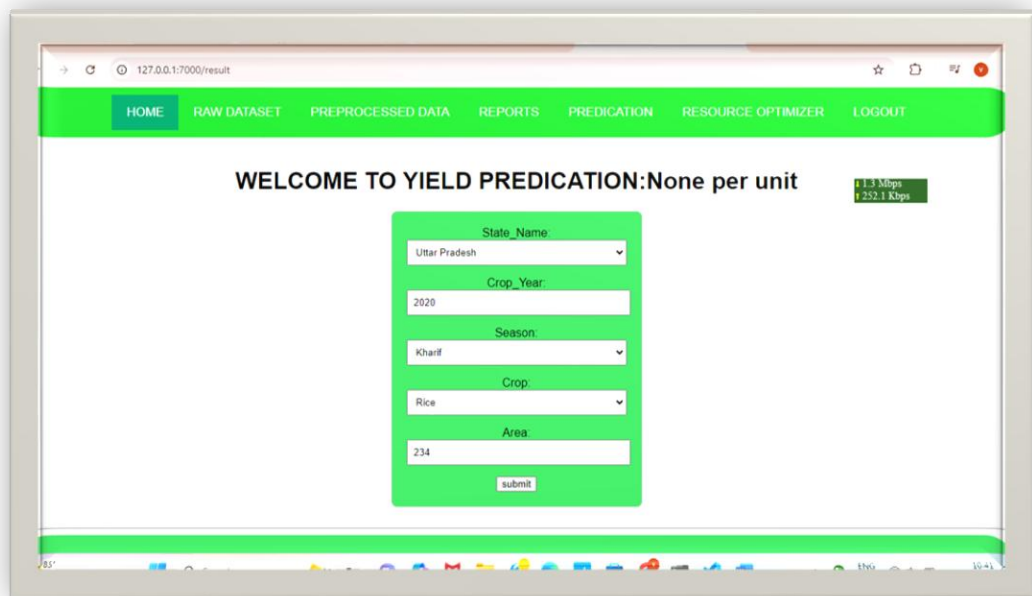


Fig 5.12 Class details

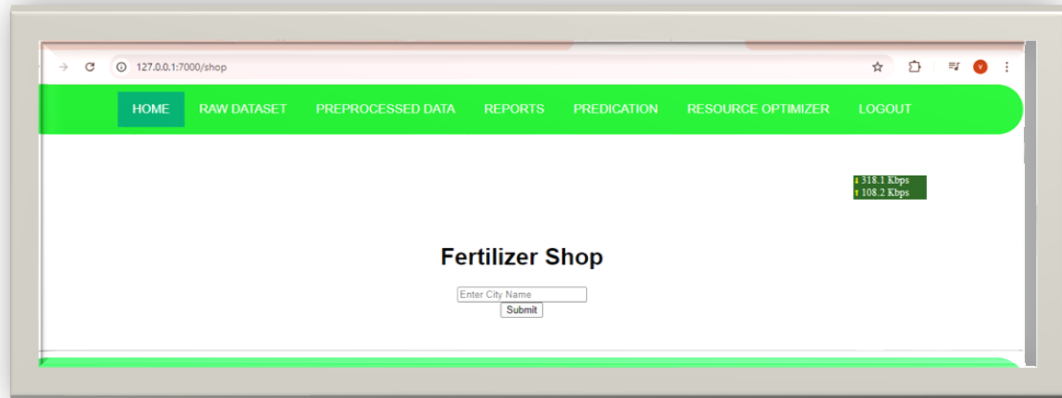
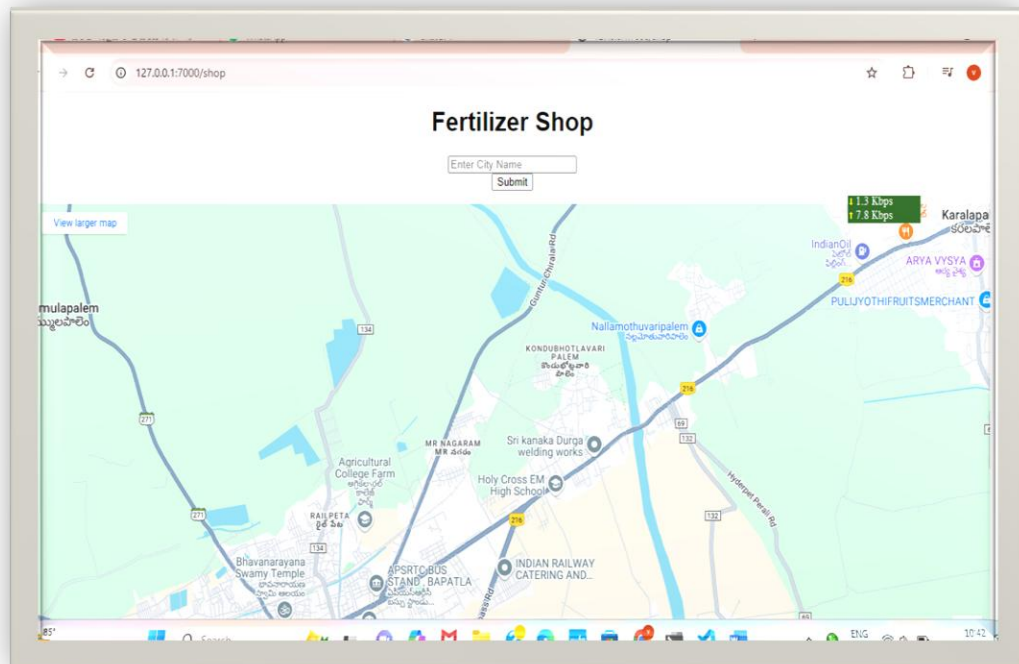


Fig 5.13 Resource Optimaziors



6. SYSTEM TESTING

6.1 INTRODUCTION TO TESTING

Software testing is a process, to evaluate the functionality of a software application with an intent to find whether the developed software met the specified requirements or not and to identify the defects to ensure that the product is defect free to produce a quality product.

Types Of Tests

1. Application Testing

It is defined as a software testing type, conducted through scripts with the motive of finding errors in software. It deals with tests for the entire application. It helps to enhance the quality of your applications while reducing costs, maximizing ROI, and saving development time.

2. System Testing

It is a level of testing that validates the complete and fully integrated software product. The purpose of a system test is to evaluate the end-to-end system specifications. Usually, the software is only one element of a larger computer-based system.

3. GUI Testing

GUI Testing is a software testing type that checks the Graphical User Interface of the Software. The purpose of Graphical User Interface (GUI) Testing is to ensure the functionalities of software application work as per specifications by checking screens and controls like menus, buttons, icons, etc.

4. Security Testing

Security Testing is a type of Software Testing that uncovers vulnerabilities of the system and determines that the data and resources of the system are protected from possible intruders. It ensures that the software system and application are free from any threats or risks that can cause a loss. Software Testing is a method to check whether the actual software product matches expected requirements and to ensure that software product is Defect free. It involves execution of software/system components using manual or automated tools to evaluate one or more properties of interest.

6.2 Test Cases

Test case writing is a major activity and considered as one of the most important parts of software testing. It is used by the testing team, development team as well as the management. If there is no documentation for an application, we can use the test case as a baseline document.

**TABLE 6.1: TEST CASES FOR ENHANCING PREDICTIVE ACCURACY
FOR AGRICULTURAL CROP YIELDS IN INDIAN STATES**

S.no	Test Case	Excepted Result	Result	Remarks(IF Fails)
1	User Register	If User registration successfully.	Pass	If already user email exist then it fails.
2	User Login	If Username and password is correct then it will getting valid page.	Pass	Un Register Users will not logged in.
3	Random forest	The request will be accepted by the random forest	Pass	The request will be accepted by the random forest otherwise its failed
4	Decision Tree	The request will be accepted by the Decision Tree	Pass	The request will be accepted by the Decision Tree otherwise its failed

5	Naive Bayes and Gradient Boosting	The request will be accepted by the random forest Naive Bayes and Gradient Boosting	Pass	The request will be accepted by the Naive Bayes and Gradient Boosting otherwise its failed
6	View dataset by user	Data set will be displayed by the user	Pass	Results not true failed
7	User predication	Display reviews with true results	Pass	Results not true failed
8	Calculate accuracy	accuracy calculated	Pass	macro avg and weighted avg not displayed failed
9	Admin login	Admin can login with his login credential. If success he get his home page	Pass	Invalid login details will not allowed here
10	Admin can activate the register users	Admin can activate the register user id	Pass	If user id not found then it won't login.

7.CONCLUSION

7.1 PROJECT CONCLUSION

In conclusion, the application of power transformations, specifically Box-Cox and Yeo-Johnson, has yielded substantial improvements in the predictive accuracy of machine learning models for forecasting agricultural crop yields in Indian states. By effectively addressing the non-normal distribution of data inherent in agricultural datasets, these transformations have significantly enhanced the performance of predictive models. The results demonstrate a consistent trend wherein models incorporating power transformations outperform those without across various metrics, with particularly notable gains observed in complex algorithms like treebased models and neural networks. Moreover, these findings hold considerable promise for stakeholders in the agricultural sector, including farmers, policymakers, and researchers. By providing more accurate and reliable forecasts of crop yields, these enhanced models can inform better decision-making processes, facilitate more efficient resource allocation, and ultimately contribute to the promotion of sustainable agricultural practices and improved food security in India. Going forward, further research in this area could explore additional avenues for optimizing predictive models and integrating real-time data sources, thereby continuing to advance the field of agricultural forecasting and its practical applications. Overall, this study underscores the transformative potential of leveraging advanced statistical techniques within machine learning frameworks to address complex challenges and drive meaningful progress in agricultural development.

7.2 FUTURE ENHANCEMENT

In Future, there should be a new proposed method to the extract pigment through feature analysis and to differentiate the type of diseases from image classification

8. REFERENCES

8.1 Journals

- [1] Mayank Champaneri, Darpan Chachpara, Chaitanya Chandvidkar and Mansing Rathod, "Crop yield prediction using machine learning", Technology, vol. 9, no. 38, 2016.
- [2] Dhivya Elavarasan and PM Vincent, "A reinforced random forest model for enhanced crop yield prediction by integrating agrarian parameters", Journal of Ambient Intelligence and Humanized Computing, vol. 12, no. 11, pp. 10009-10022, 2021.
- [3] Ranjini B Guruprasad, Kumar Saurav and Sukanya Randhawa, "Machine learning methodologies for paddy yield estimation in india: a case study", IGARSS 2019-2019 IEEE International Geoscience and Remote Sensing Symposium, pp. 7254-7257, 2019.
- [4] Raksha Jayaram and K. Deepika, "Performance collation of crop yield prediction using supervised machine learning algorithms", Journal of emerging technologies and innovative research, 2020.
- [5] Alfiya Kuznetsova, Zarema Zagirova and Zhibek Omarhanova, "Problems of poverty and motivation of workers to labour in the field of agriculture as effects of stagnant economy", 2018. 2407 Authorized licensed use limited to: Zhejiang University. Downloaded on October 24, 2024 at 04:42:02 UTC from IEEE Xplore. Restrictions apply. 2024 10th International Conference on Advanced Computing and Communication Systems (ICACCS)
- [6] Pallavi Shankarrao Mahore and Aashish A Bardekar, "Crop yield prediction using different machine learning techniques", 2021.
- [7] W Mupangwa, L Chipindu, I Nyagumbo, S Mkuhlani and G Sisito, "Evaluating machine learning algorithms for predicting maize yield under conservation agriculture in eastern and southern africa", SN Applied Sciences, vol. 2, no. 5, pp. 1-14, 2020.
- [8] James A Nichols, Hsien W Herbert Chan and Matthew AB Baker, "Machine learning: applications of artificial intelligence to imaging and diagnosis", Biophysical reviews, vol. 11, no. 1, pp. 111-118, 2019.
- [9] S. V. Patel, "Location based crop prediction using data mining and machine learning", 2020.

[10] Baisali Ghosh. A Study to Determine Yield for Crop Insurance using Precision Agriculture on an Aerial Platform. Symbiosis Institute of Geoinformatics Symbiosis International University 5th & 6th Floor, Atur Centre, Gokhale Cross Road, Model Colony, Pune – 411016.

8.2 WEB LINKS:

- <https://scikitlearn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>
- <https://lightgbm.readthedocs.io/>
- https://en.wikipedia.org/wiki/Random_forest
- https://en.wikipedia.org/wiki/Power_transform#Yeo–Johnson_transformation
- https://en.wikipedia.org/wiki/Gradient_boosting

8.3 TEXT BOOK:

- "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow"
- "Introduction to Machine Learning with Python"
- "Pattern Recognition and Machine Learning"
- "Machine Learning for Agriculture: Principles and Applications"